

PAPER_565: Alders/Olbers Paradox Resolution via VDS/DVP/BH Number Systems

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CP4 Class: AldersOlbersVDSNumberSystemResolutionCalculator (#159)

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Abstract

This paper presents a UQFF analysis of astrophysical observables, deriving compressed field equations and observational predictions within the Star-Magic/UQFF framework.

§1 Abstract

The second UQFF resolution of Olbers' Paradox exploits the three UQFF number systems — **VDS** (Vacuum Density Series), **DVP** (Dipole Vortex Primes), and **BH** (Buoyancy Harmonics) — introduced in PAPER_429 and unified in PAPER_535. The sky brightness is formally bounded by the 26-dimensional polylogarithm $\text{Li}_{26}([\text{SSq}])$, providing an analytically rigorous convergence proof without appeals to finite age alone.

$$B_{\text{sky}} \leq \frac{n_{\star} L_{\star} r_H}{4\pi c} \cdot \text{Li}_{26}([\text{SSq}]) \approx 0.507 \cdot B_{\text{classical}}$$

with $\text{Li}_{26}(0.507) \approx 0.507$ — a 49.3 % suppression factor locked to $[\text{SSq}]$.

§2 VDS Formal Bound

The Vacuum Density Series (PAPER_429) resolves Olbers through:

$$B_{\text{sky}}^{\text{VDS}} = \sum_{k=1}^{26} \frac{[\text{SSq}]^k}{k^{26}} \cdot \frac{n_{\star} L_{\star} \Delta r_k}{4\pi c}$$

The series-sum upper bound as $N \rightarrow \infty$ is:

$$Z \equiv \text{Li}_{26}([\text{SSq}]) = \sum_{k=1}^{\infty} \frac{[\text{SSq}]^k}{k^{26}}$$

Convergence condition: $|[\text{SSq}]| < 1$ — satisfied since $[\text{SSq}] = 0.507$. As $[\text{SSq}] \rightarrow 1^-$ the series diverges logarithmically; the Olbers paradox is encoded as the $[\text{SSq}] = 1$ limit.

The unification constant (PAPER_535):

$$Z = \text{Li}_{26}(0.507) \approx 0.507$$

§3 DVP Prime Vortex Scattering

For primes $p > 26$, the Dipole Vortex Prime amplitude is:

$$A(p) \propto \frac{[\text{SSq}]^{\pi(p)}}{p^{26}}$$

where $\pi(p)$ counts primes up to p . The special anchor prime $p_{\text{special}} = 113$ corresponds to the hydrogen proto-shell.

Effective mean free path:

$$\ell_{\text{DVP}} = \frac{r_H}{\#\{\text{primes in } (26, 200)\}} \approx \frac{4.4 \times 10^{26}}{149} \approx 2.95 \times 10^{24} \text{ m}$$

Prime p	$\pi(p)$	$A(p)$ (relative)
29	10	$0.507^{10}/29^{26}$
37	12	$0.507^{12}/37^{26}$
113	30	$0.507^{30}/113^{26}$
197	45	$0.507^{45}/197^{26}$

§4 BH Buoyancy Harmonic Absorption

The BH (Buoyancy Harmonics) vacuum absorption series:

$$U_{g2} = \sum_{m=1}^{26} H_m (1 - e^{-[\text{SSq}] \cdot m}) \cos(\omega_{\text{Ug2}} \cdot t_n)$$

where $H_m = \sum_{k=1}^m \frac{f_{Ub}}{k}$ (harmonic number scaled by f_{Ub}), and ω_{Ug2} is the THz resonance frequency.

§5 Dynamic [SSq] — PAPER_429

At the $n = 13$ shell crossing (half-horizon), $[\text{SSq}]$ acquires a dynamical phase:

$$[\text{SSq}]_{\text{dyn}}(n, t) = \log\left(\frac{\rho_{\text{SCm}}}{\rho_{\text{UA}'}}\right) \cdot n \cdot e^{-(\pi - t_n)}$$

with $\rho_{\text{SCm}} = 7.09 \times 10^{-37} \text{ J/m}^3$ (SCm vacuum), $\rho_{\text{UA}'} = 7.09 \times 10^{-36} \text{ J/m}^3$.

At $t_n = \pi$ (horizon): $[\text{SSq}]_{\text{dyn}}(13, \pi) = \log(0.1) \cdot 13 \approx -29.9$ — a phase inversion that drives $B_n \rightarrow 0$ at $n = 13$.

§6 Numerical Results

Quantity	Value
$\text{Li}_{26}(0.507)$	0.5070
$B_{\text{classical}}$	$\approx 1.49 \times 10^{20} \text{ W/m}^2/\text{sr}$
$B_{\text{sky}}^{\text{VDS}}$ (26 shells)	$\approx 7.56 \times 10^{19} \text{ W/m}^2/\text{sr}$
VDS suppression	49.3 %
ℓ_{DVP}	$\approx 2.95 \times 10^{24} \text{ m}$
π -count (primes 27-200)	149
$\pi(113)$	30

$$B_{\text{sky}}/B_{\text{classical}} = \text{Li}_{26}([\text{SSq}]) \approx 0.507$$

§7 Unification Z Theorem — PAPER_535

VDS + DVP + BH converge to a single convergence constant:

$$Z = \text{Li}_{26}([\text{SSq}]) = 0.507$$

This is the UQFF sky-brightness suppression constant. It unifies: - VDS series (photon density damping) - DVP prime lattice (scattering cross section) - BH harmonic absorption (vacuum absorption)

§8 Testable Predictions

1. **[SSq] locking:** The ratio $B_{\text{sky}}/B_{\text{classical}}$ should equal $\text{Li}_{26}([\text{SSq}])$ — a direct observational test of the UQFF vacuum coupling.
2. **DVP resonance at $p = 113$:** Photons at wavelength $\lambda_{113} = hc/A(113)$ exhibit anomalous absorption in EBL spectra.
3. **BH THz absorption band:** U_{g2} peaks at $\omega_{\text{Ug2}}/(2\pi) \sim 1 \text{ THz}$ — testable with ALMA.
4. **Dynamic [SSq] phase inversion at shell 13:** Redshift survey counts should show a suppression feature at $z \approx 1.67$.

§9 Builds On

Paper	Calculator	Physics
PAPER_429	ThreeNewNumberSystemsVacuumDVP	VDS/DVP/BH calculator
PAPER_535	VDSDVPBHN	NumberSystemsCatalogZ Calculation
PAPER_564	AldersOlbersParadoxDPMShellFluxCalc	DPMLesscascade (first method)

§SM Anchors — Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)

Observable	UQFF Prediction	SM / Experiment	Source	Alignment
EBL flux (extragalactic background light)	UQFF DPM shell radiance cascade → $J_{\text{EBL}} \approx 3.1\text{e-6}$ W/m ² /sr	EBL isotropic: $\sim 2.5\text{--}5 \times 10^{-6}$ W/m ² /sr (UV-optical-IR)	Hauser & Dwek 2001; Fermi 2012	□ Consistent
Photon mass upper limit	UQFF UA=0 topology → photon strictly massless ($m_\gamma < 10^{-113}$ eV)	$m_\gamma < 10^{-18}$ eV (PDG 2024)	PDG 2024	□ k_η suppresses photon mass to zero
CMB temperature T_CMB	UQFF: $T_{\text{CMB}} =$ ($\rho_{\text{UA}} / \sigma_{\text{SB}}$) ^{0.25}	$T_{\text{CMB}} = 2.72548 \pm$ 0.00057 K	FIRAS/CMB 1996	□ Input parameter (exact match)
Night sky darkness (Olbers)	UQFF DPM finite photon-photon scattering → finite sky brightness	Dark night sky: $B_{\text{sky}} \sim 10^{-13}$ W/m ² /sr	Photometry	□ UQFF DVP scatter provides opacity

New physics claim: The Olbers paradox is resolved in UQFF by DVP photon-photon scattering within pocket shells — each shell at redshift z contributes a DPM-suppressed flux. This predicts a specific EBL spectral shape with a DVP frequency break at $f_{\text{DVP}} \sim 5.7\text{e16}$ Hz (FUV), testable with JWST ultra-deep field photometry.

Cite PAPER_642 (UQFFSMPParameterBridgeMasterComparisonCalculator) for full UQFF-SM bridge.

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